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# Analysis I - Chapter 4

* I mean, technically it requires Appendix A.7. Already covered the...

## 4.1 - The Integers

1. Suppose $a-b=a+b$. Then, $a+b=a+b$. We thus cancel $b$ to obtain $a=a$, which must be true. Similarly, cancelling a given $b+b$, which by verbatim must be true. Thus, since natural numbers are well-defined, $a+b=a+b \iff a-b=a-b$. Suppose we now have an integer $c+d=a+b$. Then, $c+d=a+b$. Since natural numbers are well-defined, we have $a+b=c+d \implies a-b=c-d$.

2. Suppose $(a-b)=(a'-b')$. We must then have $0-(a-b)=0-(a'-b')$. Then, if $(a-b)$ and $(a'-b')$ are both positive natural numbers, $0+(a-b)=0+(a'-b')$. We thus get $(a-b)=(a'-b')$, which is true.

D.2.2.1 $-(a-b) = -(a'-b')$

3. Let $a=n$ $\exists n \in \mathbb{N}$. We prove $n=0$. $(-1)\times 0=-0 \implies 0\times(-1)=0-0$, whereby $0=0-0 \implies 0=0$ as discussed. Then, if for some $a$ the property is proven, we prove $(-1)\times(a+1)=-(a+1)$. Then, $(-1)\times(a+1)=-(a+1)$. Thus, $(-1)\times(a+1)=(0-(a+1))$. By definition, $(-1)$ is defined as the negation of $1=(0-1)\times((a+1)-0)=(0-(a+1))$. Then, with D.4.1.2, associativity, and $0\times n=0$, we get $((0+0)-(0+(a+1)))$. Thus we get $(0-(a+1))$, and we are done. Since negation is well-defined, we have $(-1)\times a=-a$ for some negation of $n$; namely, $a=-n$. Then, $(-1)\times a=-a$ has $(-1)\times(-n)=-(-n)$, $(0-1)\times(0-n)=-(0-n)$. Then we have $(0+n)-(0+0)=(n+0)$. Thus $(n-0)=(n-0)$ as proven.

4. Let $x=(a-b)$, $y=(c-d)$, $z=(e-f)$. Then we have $x+y=y+x \implies (a-b)+(c-d)=(c-d)+(a-b) \implies (a+c)-(b+d)=(c+a)-(d+b)$ $\implies a+c+b+d=c+a+d+b \implies a+b+c+d=c+d+a+b$, which is true. Thus, $(x+y)+z=x+(y+z) \implies ((a-b)+(c-d))+(e-f)=(a-b)+((c-d)+(e-f))$. Thus, $(a+c+b+d)+(e-f)=(a-b)+(c+e+d+f)$. Then, let $(n-0)=a+c+b+d$ $\exists n\in\mathbb{N}$ and $(m-0)=c+e+d+f$ $\exists m\in\mathbb{N}$. Then $(n+e)-(0+f)=(a+m)-(b+0)$. Then $n+e+0+f=a+m+b+0 \implies a+c+b+d+e+0+f=a+c+e+d+f+b+0$, which is true. Thus, $x+0=0+x=x \implies (a-b)+0$, whereby $0=0-0$ as discussed. Then $(a-b)+(0-0) \implies (0-0)+(a-b) \implies 0+x$. Then, $(a-b)+(0+0)$ also has $((a+0)-(b+0)) \implies (a-b)=x$. We now have $x+(-x)=(-x)+x=0$. $\implies (a-b)+(b-a) \implies (a+b)-(b+a)=(b+a)+(a-b) \implies (a+b)-(b+a)=(b+a)+(a+b)$. Thus $a+b+b+a=b+a+a+b$, which is true. We now evaluate $(a+b)-(b+a) \implies (a+b)-(a+b)$. Suppose $n=a+b$ $\exists n\in\mathbb{N}$. By definition we have $n-n=0 \implies (a+b)-(a+b)=0$. We now have $xy=yx \implies (a-b)\times(c-d)=(c-d)\times(a-b) \implies (ac+bd)-(cd+bc)=(ca+db)-(cb+da)$. Thus $ac+bd+cd+bc=ca+db+cb+da$, which is true. Thus set $y=1$, we have $x\cdot 1=1\cdot x$. Thus evaluate $1(a-b)$ whereby $(1-0)\times(a-b)=(a+0)-(b+0)$

$= (a-b)$, which gives $1x = x$. Now $x(y+z) = xy + xz \Rightarrow (a-b)[(c-b)+(d-e)]$
$= (a-b)(c-d)+(a-b)(d-e) \Rightarrow (a-b)[(c+d)-(b+e)] \Rightarrow (a(c+d)+b(b+e))-(a(b+e)+b(c+d))$
thus $(ac+ad+bb+be)-(ab+ae+bc+bd) \Rightarrow (a-b)(c-d)+(a-b)(d-e)$,
which is true. Thus, finally, $(y+z)x = yx+zx \Rightarrow x(y+z) = xy+xz$,
by preceding propositions this must be true.

5. Let $a=(x-y), b=(x'-y')\ \exists x,y,x',y' \in \mathbb{N}$. then we have
$ab = 0 \Rightarrow (x-y)\times(x'-y') = 0 \Rightarrow (xx'+yy')-(xy'+yx') = (0-0)$. Therefore
$xx'+yy'+xy'+yx' = 0+0 \Rightarrow (x+y)\times(x'+y') = 0$. this, by L.2.3.3
$(x+y) = 0$, $(x'+y') = 0$, or both. Thus, by L.2.2.2, $(x+y)=0 \Leftrightarrow x=0$ and $y=0$,
$(x'+y') = 0 \Leftrightarrow x'=0$ and $y'=0$, or both. Then, since either of these
three statements must be true, we have one of $a=(0-0)=0$, $b=(0-0)=0$,
or both, being true.

6. Suppose $ac=bc\ \exists c \in \mathbb{Z} : c \neq 0$. Then, if we subtract both sides
we have $ac-bc = 0$ by P.4.1.6. Therefore, $ac+(-bc) = 0$ which
gives $c(a+(-b)) = 0$. By P.4.1.8, either $c$, $(a+(-b))$, or both
must be zero. However, $c \neq 0$. Thus, $a+(-b) = 0 \Rightarrow a-b = 0 \Rightarrow a=b$.

7. Suppose, for contradiction, $a>b$ but $a-b$ is not a positive natural
number $\exists a,b \in \mathbb{Z}$. Then, let $a-b=0$. We therefore have
$a=b$, which by D.4.1.10 is absurd. Then, let $a-b$ be
the negation of some $n$ $\exists n \in \mathbb{N}$. Thus, $a>b$ implies
$a = b+n$ $\exists n \in \mathbb{N}$. Therefore $a-b=n$. However, $a-b$ is the negation
of some $n \in \mathbb{N}$, not $n \in \mathbb{N}$ itself. We therefore have a
contradiction. Thus, if $a>b$, let $a_2 = a+c$ and $b_2 = b+c$. We
therefore have $a_2 > b_2 \Rightarrow a+c > b+c$. The similar logic applies
to $ac > bc$. The negation of $a=b+n \Rightarrow a = n+b$ therefore implies
$-a < -b$. By axioms of transitivity, we have $a>b$ and $b>c$ giving
$a > c$. By D.4.1.10 the trichotomy $a>b$, $a<b$, or $a=b$ is
true.

(This problem sucks :|)

8. Let us prove that $\forall n \in \mathbb{Z} : n \in \mathbb{N}$. (We begin with base
case $n=0$. By A.2.1, it is known $0 \in \mathbb{N}$. Now, suppose we
prove this theorem $\exists n \in \mathbb{Z}$. Then, $n++ := n+1$ must always give
some other natural number whereby $n++ \in \mathbb{N}$. However, for
$n=-2$, we have $n++ \notin \mathbb{N}$. Thus, induction is no longer sufficient
within the integers.

A.2.2

9. We have base case $n=0$ $0\times 0 = 0$ by O.2.3.1. Thus,
$0^2 = 0 \Rightarrow 0^2 \in \mathbb{N}$. Suppose, now, that $n$ is some positive
natural number. By O.2.3.1 discussion we have $n \times n$ also be
some natural number. Thus $n^2 \in \mathbb{N}$. Lastly, suppose $n$ is the
negation of some $m$ $\exists m \in \mathbb{N}$. then we have $n^2 = n \times n$ by

defined as $n = 0 - m$. Thus $(0-m)\times(0-m) = (0+m^2)-(0+0)$. Thus $n^2 = n\times n = m^2$, which as proven, $m^2$ must be a natural number $\forall m \in \mathbb{N}$.

10. Suppose we have $a--b = \{(c,d) \in \mathbb{N}^2 : a+d = b+c\}$. Thus, we have $a--b$ be an integer iff another integer $c--d \in \mathbb{N}^2$ has $a+d = b+c$, which is colloquially equivalent to the verbatim of D.4.1.1. Thus, we create $\mathbb{Z} = \{a--b = (a,b) \in \mathbb{N}^2\}$.

## 4.2 The Rationals

1. Suppose we have a rational number $a//b$ $\exists a,b \in \mathbb{Z}$. Then, $x = a//b$ represents an object, whereby $x = x$ is established by symmetry. Now, suppose we have $y = a//b$ $\exists y \in \mathbb{Q}$. Then, if $x = y$, $a//b = a//b$, which must be true by symmetry. Then, if $x = y$, $y = x$ by transitivity. Thus, $x = y \Rightarrow ad = cb$, which gives $acd = cbd \Rightarrow a = a$. Since the integers are well defined, this holds.

2. Suppose $a//b = a'//b'$. Then we show $(a//b)(c//d) = (a'//b')(c'//d')$ $\Rightarrow (ac)//(bd) = (a'c')//(b'd') \Rightarrow acb'd = a'c'bd$. Thus, $ab' = a'b$. And by $a//b = a'//b' \Rightarrow ab' = a'b$, this must be true. Now, let $-(a//b) = (-a')//b'$. We must then have $(-a)//b = (-a')//b'$. Then $(-a)b' = (-a')b$. Recall that $ab' = a'b$. Thus, take the negation of both to get $-(ab') = -(a'b) \Rightarrow (-a)b' = (-a')b$. Our claim thus follows.

3. Let $x = a//b$, $y = c//d$ then $x+y = y+x \Rightarrow a//b + c//d = c//d + a//b$. Then $(ad+bc)//(bd) = (cb+da)//(db) \Rightarrow (ad+bc)(db) = (cb+da)(bd) = ad^2b + b^2cd = cb^2d + d^2ab$. By rearranging, we get $d^2cb + b^2cd = d^2cb + b^2cd$, thus it is true. Now let $y = 0$. We have $x+y = y+x \Rightarrow x+0 = 0+x$. To show $x = 0$, let $a//b + 0// = a//b$ by 4.4.2.3. Then $(a)(c) + (b)(0))//((b)(1)) = a//b$. Thus $(a+0)//b = a//b \Rightarrow a//b = a//b$. Now, let $y = -(a//b) = (-x)$. Thus $x+y = y+x$ gives $x+(-x) = (-x)+x$. Then, we show $x+(-x) = 0 \Rightarrow (a//b) + (-a)//b = 0//1$. Thus $(ab+(-a)b)//(b(-b)) = 0//1$. Thus, $(b(a+(-a)))//(+ab) = 0//1$, which gives $0//(-ab) = 0//1$. Assuming $-ab \neq 0$, this must be true. We now have $xy = yx \Rightarrow (a//b)(c//d) = (c//d)(a//b) \Rightarrow (ac)//(bd) = (ca)//(db)$. And therefore by associativity we have $(ac)//(bd) = (ac)//(bd)$, which must be true by the axiom of reflexivity. Suppose we have $z = (e)//(f)$. Then suppose we set $(xy)z = x(yz)$. We then have $[(a//b)(c//d)](e//f) = (a//b)[(c//d)(e//f)]$. Thus $[(ac+bc)//(bd)](e//f) = (a//b)[(cf+de)//(df)]$. Thus $[f(ac+bc)+e(bd)]//(fbd) = [a(cf+de)+b(de)]//(fbd)$. Expanding this, we get a trivial exercise to show equality. Now, suppose $y = 1$ such that $1y = yx \Rightarrow x1 = 1x$. Thus, we get $(a//b)(1//1) = (a//b)$. We then have $(a)//(b1) = (a//b) \Rightarrow (a)//(b) = (a//b)$, thus $(a//b) = (a//b)$. We now

here $x(y+z) = xy+xz$. $(a//b)(c//d)+(e//f) \Rightarrow (a//b)[(cf+de)//(df)] \Rightarrow (acf+ade)//(bdf)$, which can be seen with $xy+xz = (a//b)(c//d)+(a//b)(e//f) \Rightarrow (ac)//(bd)+(ef)//(bf)$. Thus, $(acf+ade)//(bdf)$ and we are done. $(y+z)x$ follows similar logic.

4. We show at least algebraically one of $a, b, c$ must be true. With $x = a//b$, we have 5 cases: $a=0$ & $b \neq 0$, $a<0$ & $b>0$, $a<0$ & $b<0$, $a>0$ & $b>0$, $a>0$ & $b<0$. The first gives $x=0$, fourth gives $x$ being positive by D-4.2.6, thus both second and fifth gives $x$ being negative, and third gives $x$ being positive as the negation of a negation returns back to positive rational numbers. Now, if $x=0 \Rightarrow a=0$ whereby $b \neq 0$ and $a \notin \mathbb{Z}^+$, thus it cannot be either positive or negative. If $x$ is positive, then by definition $a \in \mathbb{Z}^+$ and $b \notin \mathbb{Z}^+$, whereby there is no other $c \in \mathbb{Z}^+$ s.t. that $c = -a$, thus $x$ cannot be negative. Finally, if $x$ is negative, then $x = (-c)//b \; \exists c \in \mathbb{Z}^+$ whereby no $a = -c$ has $a \in \mathbb{Z}^+$, thus it cannot be positive.

5. Let $x = a//b$, $y = c//d$ and $z = e//f$. There is a trichotomy for both $x$ and $y$: let $\mathbb{Q}^+$ denote all positive rational numbers, and $\mathbb{Q}^-$ all negative numbers. If $P(x) = P(y)$ in relation to L.4.2.7, we may have $x=y$, $x-y \in \mathbb{Q}^+$, or $y-x \in \mathbb{Q}^+$. If $P(x) \neq P(y)$, we still have $x-y \in \mathbb{Q}^+$, or $y-x \in \mathbb{Q}^+$, the logical contrary is proven similarly. Now, if $x<y \Rightarrow x-y \in \mathbb{Q}^-$. Thus $x+(-y) \in \mathbb{Q}^- \Rightarrow (-y)+x \in \mathbb{Q}^-$. This implies $(-y)+x<0$, which may only happen when $y>x$. If $x<y$, then $x-y \in \mathbb{Q}^-$. $y<z \Rightarrow y+z \in \mathbb{Q}^-$. Thus we have $x-y = (-a)//b \; \exists a, b \in \mathbb{Z}$ and $y-z = (-c)//d \; \exists c, d \in \mathbb{Z}$. Then, $(x-y)+(y-z) \Rightarrow ((-a)//b)+((-c)//d) \Rightarrow ((-a)d+b(-c))//bd \Rightarrow (-ad-bc)//bd$. By definition, we know $bd>0$ and therefore $(-ad-bc)<0$. Thus $x-z \in \mathbb{Q}^- \Rightarrow x<z$. Now, $x<y \Rightarrow x+z<y+z$. This has $x-y$ be a negative rational number. By P.4.2.4, we have $(x+z)-(y+z) = x-y$, thus $(x+z)-(y+z)$ is also a negative number, thereby $x+z<y+z$ holds iff $(x+z)-(y+z)$ is a negative rational number finally, we have $x<y$ and $z$ being positive implying $xz<yz$. First, we know $x-y$ is negative. Similarly, $xz<yz \Rightarrow xz-yz$ is negative. By P.4.2.4, $z(x+(-y)) = xz-xy$, whereby $(x+(-y)) = x-y$, thus we know $z(x-y)$ must be negative since $z$ is positive. A positive $z$ times a negative $(x-y)$ gives a negative $z(x-y) \Rightarrow xz<yz$.

6. We have $x-y$ being negative. Then for $xz>yz$, $xz-yz$ is positive. Thus, since $z(x+(-y)) = xz-yz$, with $x+(-y) = x-y$, we have $z(x-y)$ being positive. But, $x-y$ is negative. Thus, for $z(x-y)$ to be positive, $(x-y)$ must be multiplied by some negative number. It is stated, then, that $z$ is negative and therefore $z(x-y)$ is positive. Thus $xz>yz$.

## 4.3 Absolute Value and Exponentiation

1. There are three cases: if $x$ is positive $|x| = x$ by definition and thus $|x|$ is positive. Thus, $|x| > 0$ which means $|x| \geq 0$ by D.4.2.8. Now, suppose $x = 0$. then $|x| = 0$, which means $|x| \geq 0$. Now, if $x$ is negative, $|x| := -x$ gives the negation of a negation, which as discussed previously must be positive. And, thus, $|x| > 0 \Rightarrow |x| \geq 0$. This covers the trichotomy, thus all rational numbers $x$ gives $|x| \geq 0$. Now $|x+y| \leq |x| + |y|$. Now, suppose $x = y = 0$. Then $|x+y| \leq |x| + |y| \Rightarrow |0| \leq |0| + |0| \Rightarrow 0 \leq 0$, which is true. Now, suppose $x > 0$ and $y > 0$. Then $x+y > 0$, thus $|x+y| \leq |x| + |y|$ gives $x+y \leq x+y$, which must be true. Now, if $x < 0$ and $y < 0$, then we have $|x+y| \leq (|x| + |y|) \Rightarrow |x+y| \leq -x - y$. Then, $x+y < 0$ which gives $-(x+y) = -x-y$. Thus $-(x+y) \leq -x-y$ is true. If $x < 0$ and $y > 0$. we have $|x+y| \leq -x+y$. Suppose $x+y > 0$. Then $x+y \leq -x+y$ gives either $x+y = -x+y$ or $x+y < -x+y$. The former, though, implies $x = -x$, which is absurd. Thus only $x+y < -x+y$, giving $x < -x$, which is true. Now, if $x+y$ is negative, we have $-(x+y) \leq -x+y \Rightarrow -x-y \leq -x+y$, giving $-y < y$ by similar logic. Since $y > 0$, this must be true. Similar logic holds for $x > 0$ and $y < 0$. If $y \geq |x|$, we have $y \geq 0$ by (a) and P.4.2.4. Thus, suppose $y = 0$. $0 \geq |x| \Rightarrow 0 = |x|$, which means $x = 0$, as $0 > |x|$ is absurd. Then, if $x = y = 0$, $-0 \leq 0 \leq 0$ is true, further $-|0| \leq 0 \leq |0|$ is true. Now, if $y > |x|$, then $-y < -|x|$ by E.4.2.6. to show $-y \leq x$ it suffices if $-y < x$, and by $-|x| \leq x \leq |x|$, this must be true. For $|xy| = |x||y|$, let $x = y = 0$. Then $|0| = |0||0| \Rightarrow 0 = 0 \times 0 \Rightarrow 0 = 0$, which is true. If $x > 0$ and $y > 0$, $xy > 0$ by D.4.2.9. Then $xy = |x||y| \Rightarrow xy = xy$, which is true. If $x < 0$ and $y > 0$, $xy < 0$. Then $|xy| = |(-x)y| \Rightarrow -(xy) = (-x)y$, which algebraically is true. $x > 0$ and $y < 0$ is proven similarly. If $x < 0$ and $y < 0$, $xy > 0$ and thus $|xy| = (-x)(-y) \Rightarrow xy = xy$. $d(x,y) \geq 0 \Rightarrow |x-y| \geq 0$, which by (a) is true. $d(x,y) = 0 \Rightarrow |x-y| = 0 \Rightarrow x-y = 0 \Rightarrow x = y$. Then, $d(x,y) = d(y,x)$ implies $|x-y| = |y-x|$. It can be shown $-(x-y) = y-x$. Thus, $|x-y| = |y-x|$ by (d). Lastly, $d(x,z) \leq d(x,y) + d(y,z) \Rightarrow |x-z| \leq |x-y| + |y-z|$. Then, we have $|x-y+y-z| \leq |x-y| + |y-z|$, which is true by (b). QED! ✓ this one deserves it

2. If $x = y \Rightarrow d(x,y) = 0$. Then, $0 \leq \varepsilon \ \forall \varepsilon \in \mathbb{Q}$ as $\varepsilon > 0$, by definition. Suppose the latter is false. Then, $x \neq y$ but $y$ is $\varepsilon$-close to all $x$. Then, $|x-y| > 0$ such that $x - y \leq \varepsilon \ \forall \varepsilon \in \mathbb{Q}$. Let $\varepsilon = \frac{(x-y)}{2}$. Clearly $x - y > \frac{(x-y)}{2}$, and yet $x - y \leq \varepsilon \Rightarrow x - y \leq \frac{(x-y)}{2}$, a contradiction. Now, $x$ being $\varepsilon$ close to $y$ implies $|x-y| \leq \varepsilon$, and the opposing having $|y-x| \leq \varepsilon$. It has been shown by (d) that $|x-y| = |y-x|$, thus both statements are equivalent. We have $|x-y| \leq \varepsilon$ and $|y-z| \leq \delta \ \exists \varepsilon, \delta > 0$. Then we want $|x-z| \leq (\varepsilon + \delta)$. Notice $|(x-y)+(y-z)| = |x-z|$, whereby $|(x-y)+(y-z)| \leq |x-y| + |y-z|$. Since $|x-y| \leq \varepsilon$ and $|y-z| \leq \delta$, by D.4.2.4 we have $|(x-y)+(y-z)| \leq (\varepsilon + \delta)$. $d(x,y) \leq \varepsilon$ and $d(z,w) \leq \delta \Rightarrow |x-y| \leq \varepsilon$ and $|z-w| \leq \varepsilon$

Then, $(x-z)-(y-w)=(x-y)-(z-w)$ whereby $|(x-y)-(z-w)| \le |x-y|+|(w-z)| \le \varepsilon+\delta$. Thus, $|(x-z)-(y-w)| \le \varepsilon+\delta$. I omit $x+z$ and $y+w$ as it is exactly the same proof. Now, if $\varepsilon>0$ and $|x-y| \le \varepsilon$, $\varepsilon' > \varepsilon \Rightarrow \varepsilon < \varepsilon'$ and thus $\varepsilon \le \varepsilon'$ is also true. Then, $|x-y| \le \varepsilon < \varepsilon' \Rightarrow |x-y| < \varepsilon'$. Suppose, now, that $y \le w \le z$. Then $|y-x| \le \varepsilon$ and $|z-x| \le \varepsilon$. Since $w \le z$, $w-x \le z-x$. Then, $|w-x| \le |z-x| \le \varepsilon \Rightarrow |w-x| \le \varepsilon$. If $w \le y$, then $w-x \le y-x$. whereby $|w-x| \le |y-x| \le \varepsilon$. Thus, we must have $z \le w \le y$ or $y \le w \le z$. We now want $|xz-yz| \le \varepsilon|z|$. Then, $|z(x-y)| \le \varepsilon|z|$. Now, if $|x-y| \le \varepsilon$ for $\varepsilon > 0 \Rightarrow x-y \ge 0$. Thus $x-y$ cannot be negative. We then have two scenarios: $|z(x-y)| = \varepsilon|z|$ or $|z(x-y)| < \varepsilon|z|$. For the latter, since $\varepsilon|z| > 0$, $z|x-y| = |(x-y)z| = |xy-yz|$. Thus both $xy$ and $yz$ are $\varepsilon|z|$ close.

3. $x^n x^m = x^{n+m}$. For $n=0$: $x^0 x^m = x^{0+m} \Rightarrow 1 x^m = x^m \Rightarrow x^n = x^m$. Inductively, we get $x^{n+1}x^m = x^{(n+1)+m}$. Then $[x^n x^1]x^m = x^{(n+m)+1} \Rightarrow x^n x x^m = x^{n+m}x$, and therefore $x x^n x^m = x x^n x^m$. With $(x^n)^m = x^{nm}$, let $n=0$: $(x^0)^m = x^{0m} \Rightarrow (1)^m = x^0 \Rightarrow 1^m = 1$, which is true $\forall m \in \mathbb{N}$. Then, $(x^{n+1})^m = x^{(n+1)m} \Rightarrow x^{(n+1)m} = x^{(n+1)m}$ by inductive hypothesis. With $(xy)^n = x^n y^n$, let $n=0$. Then $(xy)^0 = x^0 y^0 \Rightarrow 1 = 1 \times 1 \Rightarrow 1 = 1$. Then, $(xy)^{n+1} = x^{n+1} y^{n+1} \Rightarrow (xy)^n (xy) = x^n x y^n y$, giving $x^n y^n xy = x^n x y^n y$, inductively being true. Now, if $n \ge 0$, we start with $n=1$: $x^1 = 0 \Rightarrow x^1 \times x^0 \Rightarrow x = 0$. Now, $x^{n+1}$ has $x^n x^1 = 0 \Rightarrow x^n x = 0$. As shown, since $x=0 \Rightarrow 0^n 0 = 0$ inductively. For $x^n \ge y^n \ge 0$, let $n=0$: $x^0 \ge y^0 \ge 0 \Rightarrow 1 \ge 1 \ge 0$, which is true. Now, $x^{n+1} \ge y^{n+1} \ge 0$ gives $x^n x \ge y^n y \ge 0$. If $x \ge y \ge 0 \Rightarrow x^n \ge y^n \ge 0$, then there are two scenarios: $x=y \Rightarrow x^n = y^n \Rightarrow x^{n+1} \ge y^{n+1}$ as $x^n x = y^n y$, or $x > y \Rightarrow x \neq 0$, thereby having $x^n > y^n$. Thus, $xx^n > y^n y$ and $x^{n+1} \ge y^{n+1}$. The corollary is proven similarly. Lastly, $|x^n| = |x|^n$. For $n=0$: $|x^0| = |x|^0 \Rightarrow |1| = 1$. Since $1$ is positive, $|1| = 1$ gives $1 = 1$. Now, $|x^{n+1}| = |x^n| \Rightarrow |x^n x| = |x^n||x|$. By P.4.3.3.(d), $|x^n x| = |x^n||x|$, closing the induction.

A bit ravenous ↓

4. For all three propositions, if $n, m \in \mathbb{N}$, then P.4.3.10(c) proves these. Suppose, then, $n \in \mathbb{N}$ but $m \notin \mathbb{N}$. By the trichotomy, $m = (-a)$ $\exists a \in \mathbb{Z}^+$. Then we have $x^n x^{-a} \Rightarrow \frac{x^n}{x^a}$. Now, $x^{n-a} \Rightarrow x^{n+(a(-1))}$, thereby all of these are well-defined such that $\frac{x^n}{x^a} = x^n \cdot x^{-a} \Rightarrow \frac{x^n}{x^a}$. Now, if $(x^n)^m = x^{nm}$, suppose $n$ was negative but $m \ge 0$. $\left(\frac{1}{x^n}\right)^m \Rightarrow \frac{1^m}{x^{nm}} \Rightarrow 1 \cdot x^{-(nm)} \Rightarrow x^{-(nm)}$, which is true. Now, $(xy)^n = x^n y^n$ has not been proven if $n$ is negative. $(xy)^{-a}$ $\exists a \in \mathbb{Z}^+$. Then, $\frac{1}{(xy)^a} \Rightarrow \frac{1}{(x^a y^a)} \Rightarrow \frac{1}{x^a} \cdot \frac{1}{y^a} \Rightarrow (x)^{-a}(y)^{-a} \Rightarrow (x)^n (y)^n \Rightarrow x^n y^n$. If $n$ is positive, by P.4.3.10(c) we have $x \ge y > 0 \Rightarrow x^n \ge y^n > 0$. We induct with $n=1$: $x \ge y > 0 \Rightarrow x^1 \ge y^1 > 0$, giving $x \ge y > 0$, which by definition is true. Then, $x^{n+1} \ge y^{n+1} > 0 \Rightarrow x^n x \ge y^n y > 0$. Suppose $x=y$: $x^n = y^n$ such that $x^n x = y^n y$, and if $x > y$, $x^n > y^n$ and therefore $x^n x > y^n y$. Now, if $n$ is negative, $0 < x^n \le y^n$ has $0 < \frac{1}{x^n} \le \frac{1}{y^n}$. $\exists a \in \mathbb{Z}^+$. then, if $x=y \Rightarrow x^{-a} = y^{-a}$ and

thus $x^n = y^n$. If $x > y$, then suppose $\frac{1}{x^a} \neq \frac{1}{y^a}$. We cannot have $x = y$ by definition, so then $\frac{1}{x^a} > \frac{1}{y^a}$. If $a = 1$: $\frac{1}{x} > \frac{1}{y}$, which cannot be true. Inductively, this absurdity holds. Thus $\frac{1}{x^a} < \frac{1}{y^a} \Rightarrow 0 < x^n \leq y^n$. Now, suppose both $x$ and $y$ are positive. $x^n = y^n$ has $n > 0$: Suppose $x \neq y$, then $x' = x$ and $y' = y$ has $x' \neq y'$. However, $x^n = y^n \ \forall n \in \mathbb{Z}^+$, a contradiction. Now, if $n < 0$, we have $\frac{1}{x^a} = \frac{1}{y^a} \Rightarrow x = y \ \exists a \in \mathbb{Z}^+$. If $x \neq y$, then $\frac{1}{x} \neq \frac{1}{y}$. However, since $x = x'$ and $y = y'$, $\frac{1}{x'} \neq \frac{1}{y'} \Rightarrow \frac{1}{x^a} = \frac{1}{y^a} \ \exists a \in \mathbb{Z}^+$ such that $n = -a \ \forall n \in \mathbb{Z} : n < 0$. We therefore have a contradiction. Lastly, $|x^n| = |x|^n$. If $n \geq 0$, P. 4.3.10(d) proves this. Then, if $n < 0$: $\left|\frac{1}{x^a}\right| = \frac{1}{|x|^a}$. $\forall n \in \mathbb{Z} : n = -a$. Then, we use P. 4.3.3(d) to show $\left|\frac{1}{x^a}\right| = \frac{|1|}{|x^a|} = \frac{1}{|x^a|}$. Thus $\frac{1}{|x^a|} = \frac{1}{|x|^a} \Rightarrow |x^a| = |x|^a$, which has been proven for all natural numbers, whereby $a \in \mathbb{Z}^+ \in \mathbb{N}$.

5. We start with $N = 1$: $2^1 \geq 1 \Rightarrow 2 \geq 1$, which is true. Then $2^{N+1} \geq N+1$ has $2^N \cdot 2 \geq N+1$. We then have $2^N \geq \frac{(N+1)}{2}$. Clearly, since $2^N \geq N$, we want to show $2^N \geq N \geq \frac{(N+1)}{2} \ \forall N \in \mathbb{Z}^+$. Then, if $N = 1$: $1 \geq \frac{(1+1)}{2}$ has $1 \geq 1$. For $N+1$: $N+1 \geq \frac{((N+1)+1)}{2}$, thus $N+1 \geq \frac{N+2}{2} \Rightarrow N+1 \geq \frac{N}{2} + \frac{2}{2} \Rightarrow N+1 \geq \frac{N}{2} + 1$, thus $N \geq \frac{N}{2}$, which is true for all $N$. Then, $2^N \geq N \geq \frac{(N+1)}{2} \Rightarrow 2^N \geq \frac{(N+1)}{2}$, giving $2^{N+1} \geq N+1$.

## 4.4 Gaps in the rational number.

1. Suppose we have some $x$ be a rational number. If $n = x$, then we know that by the trichotomy $x$ has three cases: If $x = 0$, $n \leq x < n+1 \Rightarrow 0 \leq 0 < 1$. then if $x > 0$, $x$ must have a successor $n+1$ such that A.2.2 is satisfied. Then, if $x < 0$, $n$ must be the negation of some $a \in \mathbb{Z}^+$ such that $-a \leq x < -a+1 \Rightarrow -a \leq n < -a+1$, which must be true. Suppose, now, that $x \notin \mathbb{Z}$. Then, for $n \leq x < n+1$ to hold we must have $d(x,n) < 1$. I.e., $d(n, n+1) = 1$, whereby $n \neq x$ and $n+1 \neq x$ as $x \notin \mathbb{Z}$. Then, we also have $d(n+1, n) = 1$, such that $n < x < n+1 \Rightarrow x$ is between $n$ and $n+1$, therefore $d(n, x) < 1$. However, $|n - x| \neq 1$ and $|x - n| \neq 1$. Thus $d(n, x) < 1$. Therefore, $|n+1| > d(n, x) \ \forall n \in \mathbb{Z}$ such that there will always be some $n+1$ greater than any non-integer rational number. Then, set $N = n+1$ such that the last part of the proposition applies, whereby $n > 0$ gives $n+1 > d(n, x)$, thus a new natural number is found.

2. (a) Assume, for contradiction, that we may form an infinite descent of natural numbers. Then, $a_n \geq 0 \ \forall n \in \mathbb{N}$. If for some $k$ this is true, then we show for $k+1$: $a_n \geq k+1 \Rightarrow 0 \geq k+1$. However, axiomatically $0 \neq k+1 \ \forall k+1 \in \mathbb{N}$. Thus $a_n \neq 0$. Then $a_n > 0$. Eventually, we shall reach some $a_n = 1$, such that there is no longer any $a_n > a_{n+1} > 0$. Since, $1 > a_{n+1} > 0$

is absurd. For (b), let us work on $\mathbb{Z}$ first. Suppose $a_0 = k$ such that $a_n < a_{n+1}$. Define $a_{n+1} := a_n - 1$. Then, for all $a_n$ we have $a_n = a_{n+1} + 1$, and this $a_n < a_{n+1}$. For rational numbers, suppose $a_0 = k$. Define $a_{n+1} = \frac{a_n}{2}$. Then, for all $a_n$ we have $a_n = 2a_{n+1} \Rightarrow a_n > a_{n+1}$ such that $k \neq 0$. Then, $a_n > a_{n+1}$ will always be true.

3. Suppose a natural number $p$ is both odd and even. Then $p = 2k$ and $p = 2k+1$ $\exists k \in \mathbb{N}$. Then, $2k = 2k+1 \Rightarrow p = p+1$. However, by A.2.4 this is absurd. Suppose that $p$ is odd, we show $p^2$ is odd: $(2k+1)^2 = 4k^2 + 4k + 1$. Since $4k^2 + 4k \Rightarrow 4k(k+1)$, we show $4k(k+1) = 2k(2k+2)$ such that $2[k(2k+2)]$. Thus $4k(k+1)$ must be even. Thus, there is some $j = k(2k+2)$ $\exists j \in \mathbb{N}$, whereby $2j+1 = 4k^2 + 4k + 1$ is odd. Thus, if $p$ is odd then $p^2$ must also be odd. Given $p^2 = 2q^2$, we have $q < p$. Suppose, for contradiction, that $q \geq p$. If $q = p$, $p^2 = q^2$, such that $q^2 = 2q^2$, which implies $1 = 2$, which is absurd. Then if $q > p$, $q^2 > pq$. $2q^2 > 2pq \Rightarrow p^2 > 2pq$. Thus $p > 2q$, and yet $p < q$. We then have $2q < p < q \Rightarrow 2q < q$, which is absurd. Q.E.D.

good ends

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